

Review article

A review of clinical studies of electrodermal activity and transcranial magnetic stimulation

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ABSTRACT

There is a growing body of evidence indicative of changes in autonomic nervous system (ANS) activity in patients with disorders of the central nervous system (CNS). Non-invasive measures of the ANS, including heart rate variability (HRV), electrodermal activity (EDA), and pupillary light reflex (PLR) may have value as markers of symptom severity, subtype, risk profile, and/or treatment response. In this paper we provide an introduction into the anatomy and physiology of EDA and review the literature published after 2007 in which EDA was an outcome measure of cortical stimulation with transcranial magnetic stimulation (TMS). Eleven studies were included and considered regarding the potential of EDA as an outcome measure reflecting ANS activity in TMS research and treatment. These studies are summarized according to study population, experimental methodology, cortical region targeted, and correlation with other measures of ANS activity. Results indicate that EDA changes vary with the frequency and target of TMS. Inhibitory TMS to the dorsolateral prefrontal cortex (dlPFC) was the most common paradigm in these studies, consistently resulting in decreased EDA.

1. Introduction

Peripheral measures of the activity of the ANS, as measured by HRV and EDA, have been utilized extensively as indicators of a broad spectrum of cognitive/emotional activity. While HRV is reflective of both the sympathetic and parasympathetic systems, EDA is felt to be under sympathetic control only. An extensive literature spanning 50 years demonstrates an association between EDA and mental states, including depression and suicide (Sarchiapone et al., 2018). And while recent work suggests the potential of EDA as a biomarker in the prediction of antidepressant response (Pineau et al., 2022), there are many caveats, not the least of which is that changes in EDA reflect a small part of a complex series of ANS mediated responses (Dawson M.E., 2016)

Emerging neurostimulation therapeutics, such as TMS, modulate the CNS broadly, triggering a cascade of changes ultimately resulting in clinical improvement (O'Reardon et al., 2007). There is growing evidence to suggest that TMS stimulation of cortical regions, the dlPFC in particular, produces measurable changes in ANS activity (Iseger et al., 2020). Physiological responses provide the opportunity to track

psychological events in real time. We did not find any review articles on the effects of TMS on EDA even though the use of both methods simultaneously is becoming more common in the study of CNS disorders, prompting this inquiry.

1.1. What is EDA and how is it measured?

Sweat glands in the skin are innervated by sympathetic neurons of the ANS. Changes in skin conductance (SC) are a proven measure of the output level of sympathetic activity, reflecting an organism's response to CNS cognitive or emotional states. Primates are thought to have evolved hand perspiration along with fingerprint ridges to facilitate grip strength in demanding manual conditions (Critchley, 2002). Typically, changes in SC are measured using silver chloride electrodes and sodium chloride electrolyte gel placed on the volar surfaces of the middle phalanges of the second and third digits (Fig. 1), although the thenar and hypothenar regions of the palms or other areas of the body can be used. Connectivity between electrodes is increased as sweat, containing water and electrolytes, lowers the electrical resistance of the skin. Measurements are

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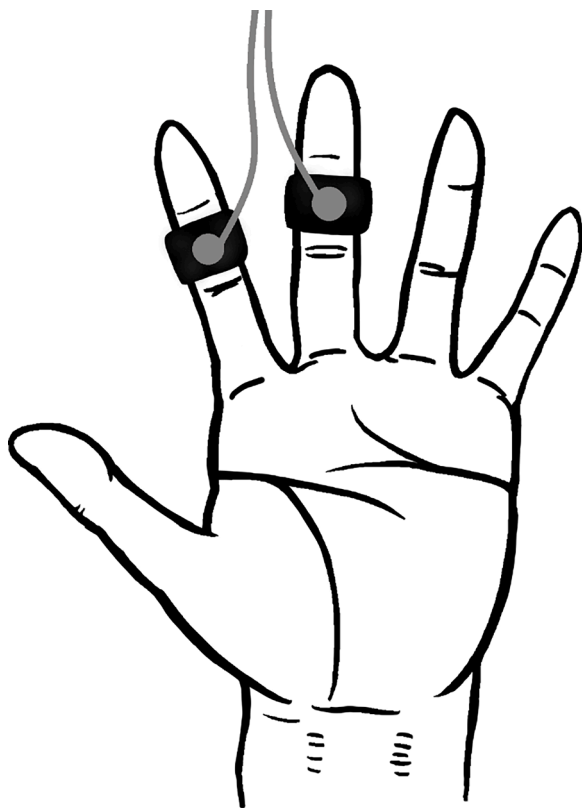


Fig. 1. EDA measurement.
This figure depicts the most common location for EDA recording. Electrodes are placed on the volar side of the middle phalanges of the second and third digits of the hand. One electrode is inactive, serving as a ground, while the other is active. Collection is signaled and recorded through wires which communicate with a computer or other recording device. EDA = electrodermal activity.

taken at a high sampling frequency to approximate continuous monitoring and are filtered and processed based on the needs of the research question.

Background levels of SC as well as fluctuations in SC in response to stimuli can be measured in this fashion, and the term EDA is applied to both. In addition, EDA encompasses all electrical properties of the skin, both tonic and phasic (Boucsein, 2012). The term galvanic skin response (GSR) has fallen out of favor when referring to changes in SC, partly because it only implies changes elicited by an external stimulus and not spontaneous fluctuations or changes with a psychological origin (Boucsein, 2012). As a result, GSR has been largely replaced with the term EDA. Terms ending in “response” (i.e., SCR, EDR) refer to phasic

Table 1
EDA abbreviations.
Common abbreviations of different EDA measurements are adapted from *Electrodermal Activity* (Boucsein, 2012). Exosomatic methods of recording apply either direct or indirect current to the skin while endosomatic recording does not apply a current and only measures intrinsic skin potential differences (Boucsein, 2012). Measurements taken with alternating current were excluded due to their rare use. Other features of SC can be tacked onto the end of EDA terms, such as “frequency” (i.e., SCR freq) or “latency” (i.e., SCR lat) with the exception of “nonspecific response” being a prefix (i.e., NS.SCR).

Recording Method	Endosomatic	Exosomatic (DC)	
	Skin potential	Skin resistance	Skin conductance
General	SP	SR	SC
Level (tonic)	SPL	SRL	SCL
Response (phasic)	SPR	SRR	SCR

EDA = electrodermal activity; SC = skin conductance.

activity occurring due to a stimulus (Table 1). Phasic activity can occur on top of tonic activity (Grace, 1991). Tonic activity, indicated by the word “level” at the end of terms (i.e., SCL), refers to the steady background activity of sweat glands and their afferent innervating neurons (Boucsein, 2012). In general, increases in EDA are taken to represent primarily the activity of the sympathetic nervous system (SNS), and increases in EDA reflect global increases in physiologic arousal.

1.2. Anatomy and CNS control of EDA

In order to understand how EDA may be influenced by cortical stimulation, some understanding of CNS control of EDA is needed. The sympathetic response is widespread throughout the organism, comprising multiple end organs, including the pupils, heart, blood vessels, lungs, gastrointestinal, and urinary systems, in response to stress or other homeostatic demands. Correspondingly, the brain structures that exert excitatory and inhibitory influences on EDA are both widespread and hierarchical, involving subcortical structures and cortical regions. Benarrach posited a central autonomic network (CAN) composed of interlinked regions of the telencephalon, diencephalon, and brainstem that serve to process afferent visceral inputs and to control the ANS via preganglionic neurons in the brainstem vagal nuclei and spinal cord (Benarroch, 1993). Our knowledge of these neural controls is derived from both human and animal studies, including lesion studies (Tranel and Damasio, 1989; Tranel, 1983; Zahn et al., 1999b) as well as limited direct electrical stimulation (Mangina and Beuzeron-Mangina, 1996) and neuroimaging studies (Fredrikson et al., 1998). Animal studies confirm that, at its most basic level, physiologic arousal reflected in the ANS via EDA can be elicited via stimulation of the brainstem reticular formation (Roy et al., 1977). At an intermediate subcortical level, thermoregulatory control is thought to be governed principally by the ipsilateral hypothalamus, with the amygdala generating excitatory and the hippocampus generating inhibitory inputs via secondary reflex arcs (Dawson M.E., 2016; Tranel, 2000). Several cortical pathways also influence central EDA activity including excitatory control via the premotor cortex and basal ganglia. The structural components of the CAN are theorized to include the ventromedial, insular, and anterior cingulate cortices, the extended amygdala including the central nucleus and bed nucleus of the stria terminalis along with the hypothalamus, and the periaqueductal gray along with brainstem nuclei including the parabrachial nucleus, nucleus of the tractus solitarius, nucleus ambiguus, and ventral medullary regions (Thayer and Lane, 2000). Most importantly, as a basis for clinical correlation with common TMS targets and connectivity, lesions of the motor association cortex including dorsolateral, orbitofrontal, and ventromedial regions of the frontal lobe as well as the anterior cingulate and supplementary motor areas have been shown to influence EDA activity (Tranel and Damasio, 1994; Zahn et al., 1999b) (Fig. 2).

EDA terminology and methodology has suffered in the past from a lack of standardization, but the field is rapidly gaining in sophistication (Posada-Quintero and Chon, 2020). This review is focused on the use of this methodology in studies of TMS stimulation targeting and on correlations with other measures of ANS activity, with the ultimate goal of assessing the potential value of EDA as a marker in TMS research and treatment.

2. Methods

We searched PubMed, Google Scholar, and Web of Science to find studies using both TMS as an intervention and EDA as an outcome. Because of the varying jargon surrounding EDA, we paired TMS with different common terms for EDA measurement as follows: “TMS and EDA,” “TMS and electrodermal activity,” “TMS and GSR,” “TMS and skin conductance.” Studies were excluded if they were duplicates, if they were not in English, if they did not pertain to TMS or EDA, or if they were done prior to 2007. This cutoff was chosen to account for advances

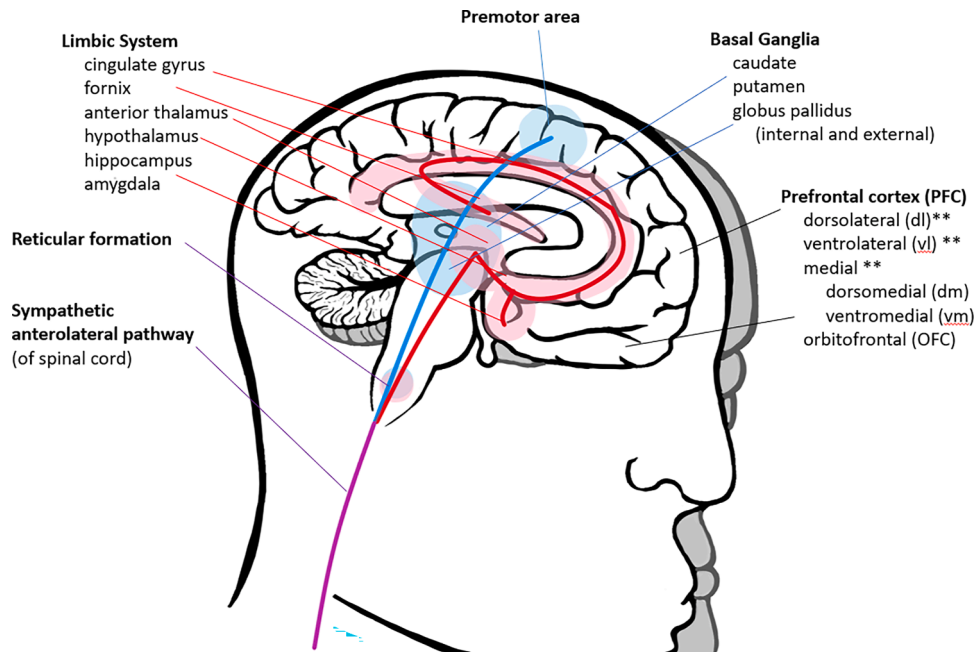


Fig. 2. EDA neural pathway.
EDA in the skin can be stimulated by multiple pathways of CNS origin. Classically, a signal originates in the premotor cortex and basal ganglia then travels down the sympathetic anterolateral spinal cord pathway to the skin (Boucsein, 2012). EDA in response to emotional stimuli arises in the limbic system, including the hypothalamus, and also leaves in the sympathetic anterolateral pathway (Boucsein, 2012). Additionally, a signal can be generated in the reticular formation of the medulla (Boucsein, 2012). Other cortical areas are implicated in EDA. Damage to the dlPFC and to the anterior cingulate cortices are associated with decreased EDA response (Critchley, 2002). The injured dlPFC is specifically associated with a diminished EDA in the setting of response to stimuli but not in motor behavior (Critchley, 2002). The vmPFC is associated with emotion and ANS control (Critchley, 2002), and damage to the vmPFC is associated with a lack of the typical EDA seen in risky decision making (Dawson M.E., 2016). The OFC is also associated with EDA in response to risk taking, and the OFC and the limbic system both function in the response emotional stimuli (Dawson M.E., 2016). In addition to being part of the limbic system response to emotional stimuli, the hypothalamus serves as the body's thermostat and as such is responsible for EDA for thermoregulation. Brain areas stimulated in included studies but not represented in the classic EDA pathway include: M1*, SMA**, and TPJ* *Indicates where a study has attempted TMS stimulation **Indicates where at least one study has shown TMS-related changes in EDA = electrodermal activity; CNS = central nervous system; dlPFC = dorsolateral prefrontal cortex; vmPFC = ventromedial prefrontal cortex; ANS = autonomic nervous system; OFC = orbitofrontal cortex; TMS = transcranial magnetic stimulation; M1 = primary motor cortex; SMA = supplementary motor cortex; TPJ = temporoparietal lobe junction.

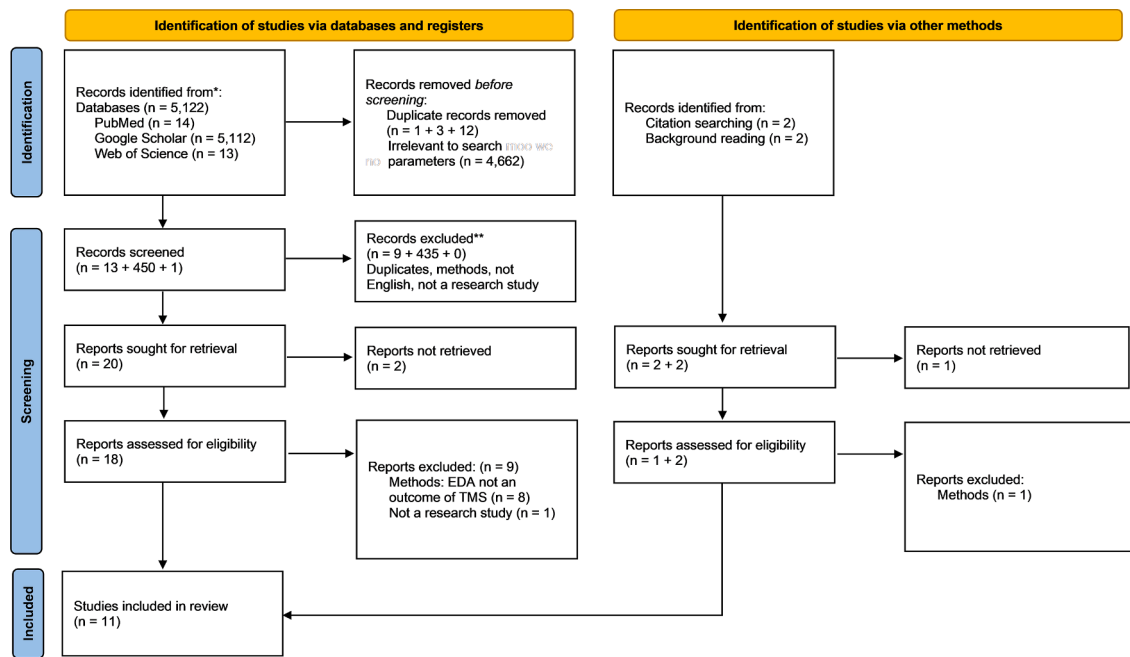


Fig. 3. Flow diagram of study selection.
This table details the sources of the studies included in our review. The process is enumerated in the methods section. The template is from the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA)(Page et al., 2021).

in TMS over time and because TMS was approved by the US Food and Drug Administration to treat depression in 2008.

In PubMed, the searches yielded 14 studies when filtered to exclude books and documents (Fig. 3). There was one duplicate study, and nine studies were excluded because their methods did not incorporate both TMS and EDA. The same search terms were also applied to the database Web of Science. Only one new study was found that was not a duplicate from PubMed. In Google Scholar, the search results were much broader and non-specific to the inputs. Searching with the terms “TMS” and “EDA” produced over 8000 results, but most of the studies contained the acronym “EDA” for a meaning other than electrodermal activity, so we searched again using “TMS” and “electrodermal activity” for more specificity. After searching for “TMS” with “GSR” then with “skin conductance,” there were a total of 5112 studies between the three Google Scholar searches. We assessed approximately the first 150 studies of each search, stopping after several consecutive pages of search results were inapplicable to the intent of the search terms since the search results became increasingly irrelevant as the list progressed. Initially, 230 studies of the approximately 450 were excluded due to apparent lack of information on TMS and/or EDA. The remaining studies were reviewed to assess if EDA was collected as an outcome of TMS, leaving only 20 studies between PubMed, Google Scholar, and Web of Science to be assessed by full-text review.

Of the 20 studies for full-text review, two were excluded because the study was not available online. Eight others were excluded because, upon further inspection, EDA was not collected as an outcome of TMS. One study was a letter to the editor and was excluded due to sparse information.

Two more studies of interest were found from citations in identified studies, another two studies were identified from previous readings on this topic. One of these studies was not accessible online, and one did not contain sufficient methods to be incorporated into the final table. In total, 11 studies were included for assessment of patterns in EDA following TMS.

We organized the included studies into a table with the EDA method, TMS method, study population, notable study outcomes, and other ANS-related data collected. We determined that the existing TMS/EDA literature is adequate to support a semi-structured review to illuminate commonalities in the available qualitative information. We focused upon experimental methodology, cortical region targeted, and correlation with other measures of ANS activity such as HRV, in order to get a look at how ANS activity as a whole responded to TMS stimulation in these studies.

3. Results

A total of eleven studies were included in this semi-structured review (Table 2). Across all studies, a total of 378 participants, representing both healthy and clinical populations, were assessed. Five of the eleven included studies had sample sizes over thirty (Berger et al., 2017; Guhn et al., 2014; Notzon et al., 2015; Tsang et al., 2009; Wang et al., 2016), five studies included a group receiving sham TMS (Berger et al., 2017; Guhn et al., 2014; Kuppawamy et al., 2008; Notzon et al., 2015; Tsang et al., 2009), and two were abstracts (Fedor et al., 2016; Kuppawamy et al., 2008). For TMS methodology, the majority used a standard figure-eight coil with a diameter of 70 mm (Bracco et al., 2017; Casanova et al., 2014; Jay et al., 2014; Kuppawamy et al., 2008; Rodigari and Oliveri, 2014; Wang et al., 2016). Most studies used low frequency (LF) TMS (Berger et al., 2017; Bracco et al., 2017; Casanova et al., 2014; Jay et al., 2014; Kuppawamy et al., 2008; Rodigari and Oliveri, 2014; Tsang et al., 2009; Wang et al., 2016), varying from 0.5 Hz to 1 Hz, while relatively few used high frequency (HF) TMS (Bracco et al., 2017; Guhn et al., 2014; Notzon et al., 2015).

Test populations varied across each study. A majority included healthy volunteers (Berger et al., 2017; Bracco et al., 2017; Guhn et al., 2014; Kuppawamy et al., 2008; Rodigari and Oliveri, 2014; Tsang

et al., 2009), two studying females only (Berger et al., 2017; Rodigari and Oliveri, 2014). Clinical populations included those with Autism Spectrum Disorder (ASD) (Casanova et al., 2014; Wang et al., 2016), medication-resistant depersonalization disorder (DPD) (Jay et al., 2014), spider phobia (Notzon et al., 2015), and non-psychotic major depressive disorder (MDD) (Fedor et al., 2016). Two clinically-focused studies included a healthy control group (Jay et al., 2014; Notzon et al., 2015). Several studies (Berger et al., 2017; Casanova et al., 2014; Fedor et al., 2016; Guhn et al., 2014; Jay et al., 2014; Notzon et al., 2015; Tsang et al., 2009; Wang et al., 2016) utilized patient-reported scales as clinically-focused assessments for affect, depression, anxiety, repetitive behavior, discomfort, and more. Clinical populations are likely to have baseline ANS perturbations. For example, a blunted SNS is associated with DPD as these patients complain of altered present moment awareness and often have muted emotional reactions, while anxiety-related disorders are associated with hyperarousal (Reutens et al., 2010; Sierra et al., 2006).

Importantly, baseline EDA variability was recorded across all populations. Two studies (Casanova et al., 2014; Wang et al., 2016) found that patients with ASD had higher baseline EDA levels with less variability to alarming stimuli. One study saw patients with DPD had lower maximum EDA levels pre-TMS treatment with less fluctuations in EDA in response to visual stimuli (Jay et al., 2014). Another study documented that patients with spider phobia had higher baseline EDA levels when exposed to spider images compared to healthy controls (Notzon et al., 2015). Patients with MDD had higher baseline EDA levels on their right wrists compared to their left wrists before TMS treatments (Fedor et al., 2016).

3.1. EDA methodology

Included studies either placed EDA sensors on the thenar and hypothenar eminences (Berger et al., 2017; Bracco et al., 2017; Guhn et al., 2014; Kuppawamy et al., 2008; Notzon et al., 2015) or on the distal index and middle fingers (Casanova et al., 2014; Jay et al., 2014; Tsang et al., 2009; Wang et al., 2016), with most studies using the nondominant hand. One study tested EDA on bilateral wrists in order to assess asymmetry (Fedor et al., 2016). Most studies began recording EDA between five and twenty minutes before until five to twenty minutes after TMS treatment. Recording EDA levels before TMS treatment was important for calibrating the devices, determining the baseline, and controlling for fluctuations in EDA that could potentially be a response to the TMS apparatus itself (Wang et al., 2016). Different programs were used for data analysis, but each method continuously monitored EDA levels during measurement time periods, filtered out fluctuations deemed insignificant, and averaged EDA levels. Fluctuations were considered notable if they exceeded a predetermined threshold or if they were temporally associated with certain stimuli, as these were assumed to be responses (Jay et al., 2014; Wang et al., 2016). Fluctuations associated with the audible clicks from the TMS machine were filtered out (Wang et al., 2016).

Many studies used a provoking factor before and/or after TMS treatment sessions to determine EDA and other ANS fluctuations as well as subjective responses to certain triggers (Berger et al., 2017; Jay et al., 2014; Kuppawamy et al., 2008; Notzon et al., 2015; Rodigari and Oliveri, 2014). Provoking factors included emotionally triggering images (Berger et al., 2017; Guhn et al., 2014; Jay et al., 2014; Rodigari and Oliveri, 2014) or physical stimuli (Kuppawamy et al., 2008). One study used a pictorial scale to rate arousal to visual stimuli (Jay et al., 2014). Another used virtual reality in order to provoke participants with a spider phobia while recording EDA response and participants' real-time anxiety and disgust (Notzon et al., 2015). The complexity of EDA dynamic responses to provoking factors is illustrated by Berger et al. who noted a simultaneous increase in EDA in response to salient visual stimuli, possibly indicating sympathetic-mediated attention capture, and a phasic heart rate (HR) deceleration, presumably indicating

Table 2**Table of Included Studies.**

This table displays all included studies by population, TMS methodology, EDA methodology, other autonomic metrics collected, and notable study outcomes.

Study	Population studied	Sample size	Brain region targeted	TMS methodology	EDA outcome	Other metrics included	other notable observations
Casanova et al., 2014	Children with ASD	18	dIPFC (6 consecutive treatments on left, right, and bilaterally)	0.5 Hz 90% MT 160 pulses per day Once per week	Reduction in EDA over course of treatment	Increase in HRV	Behavioral checklist: decreased irritability, hyperactivity, and stereotyped/compulsive behaviors
Wang et al., 2016	Children with ASD	33	dIPFC 6 X Right 6 X Left	0.5 Hz 90% MT 12 sessions/160 pulses per day/ Once per week	Reduction of EDA over course of treatment	Increase in HRV	Behavior checklist: decreased irritability, hyperactivity, and stereotyped/compulsive behaviors. EDA changes correlated with behavioral outcomes
Kuppuswamy et al., 2008	Healthy participants	4	Right dIPFC, right primary motor cortex, left cerebellum 48 h between single sessions	1 Hz 90% MT 900 pulses Sham	Latency lengthening in EDA in dIPFC group	N/A	N/A
Berger et al., 2017	Healthy females	40	Right dIPFC	1 Hz and 10 Hz 110% MT 900 pulses Sham	No statistically significant differences noted in EDA	Greater decrease in HR in 1 Hz TMS group when shown negative pictures compared to HFTMS group	N/A
Notzon et al., 2015	Spider phobia and healthy controls	83	Left dIPFC	iTBS 80% MT 600 pulses Sham	No statistically significant differences post iTBS, but phobic group was noted to have higher EDA levels during VR spider scenario	Marginally significant increase in LF and decrease in HF HRV after iTBS	Change in HRV from iTBS could indicate increase in SNS activity
Guhn et al., 2014	Healthy participants, fear conditioning and extinction paradigm	88	mPFC Side not specified	10 Hz 110% MT 1560 pulses Sham	No statistically significant differences noted	N/A	TMS group had lower arousal during extinction learning compared to sham TMS group
Bracco et al., 2017	Healthy participants	10	M1 Right or left One week between sessions	Paired pulse 100% 110% 130% 150%	Slope of the IO curve is directly related to EDA on right, inversely related to EDA on the Left M1	N/A	Right hemisphere has more SNS involvement
Rodigari and Oliveri 2014	Healthy females	18	SMA Side not specified	1 Hz 90% MT 600 pulses	EDA increased after TMS	N/A	SMA might be involved in emotional processing
Tsang et al., 2009	Healthy participants	32	Bilateral TPJ	0.5 Hz Sham	No statistically significant differences noted	No statistically significant different noted in HR	Improved mood and vigor in participants receiving TMS compared to sham
Jay et al., 2014	DPD and healthy controls	43	Right TPJ and right vIPFC	1 Hz 110% MT 900 pulses	TPJ group saw no EDA changes, but vIPFC saw increase in EDA	N/A	DPD symptoms improved after TMS to TPJ and vIPFC More pronounced EDA increases over vIPFC in DPD but over TPJ in healthy controls
Fedor et al., 2016	Depressive Episode without psychotic features	9	Not specified, but implied standard Left DLPFC 36–72 sessions of variable duration	Not specified but implied standard 10 Hz	Increased EDA on right wrist correlated with worsening depression	N/A	Asymmetry of EDA between wrists could indicate depression

ASD = autism spectrum disorder; DPD = depersonalization disorder; dIPFC = dorsolateral prefrontal cortex; vIPFC = ventrolateral prefrontal cortex; mPFC = medial prefrontal cortex; M1 = primary motor cortex; TPJ = temporoparietal lobe junction; SMA = supplementary motor cortex; MT = motor threshold; rTMS = repetitive transcranial magnetic stimulation; iTBS = intermittent theta burst; EDA = electrodermal activity; HR = heart rate; HRV = heart rate variability; HF = high frequency HRV domain; LF = low frequency HRV domain; I/O = input-output; VR = virtual reality; SNS = sympathetic nervous system.

parasympathetically mediated continued sensory processing ([Berger et al., 2017](#)). This suggests that the temporal relationship of EDA measurement relative to both provoking factors and to TMS stimulus is of considerable importance in interpreting results and therefore must be recorded. Some studies assessed changes in EDA levels over the course of multiple TMS treatments ([Casanova et al., 2014](#); [Tsang et al., 2009](#); [Wang et al., 2016](#)) while others assessed immediate EDA changes during and after a single TMS treatment ([Berger et al., 2017](#); [Bracco et al., 2017](#);

[Guhn et al., 2014](#); [Jay et al., 2014](#); [Kuppuswamy et al., 2008](#); [Notzon et al., 2015](#); [Rodigari and Oliveri, 2014](#)).

3.2. TMS methodology

Stimulation of the motor cortex (M1) was done to determine optimal TMS location and treatment intensity. In each study, motor evoked potentials were measured with electromyography, using either the

abductor pollicis brevis (Berger et al., 2017; Casanova et al., 2014; Jay et al., 2014) or the first dorsal interosseus (Kuppuswamy et al., 2008; Wang et al., 2016) muscles, to quantify the muscle activity that resulted from the TMS stimulation. The resting motor threshold (RMT) was then used to determine the intensity of TMS treatment. Studies set TMS intensity as a percentage of the RMT value. One study employed a very low intensity, 1 microTesla, magnetic field for a LFTMS paradigm (Tsang et al., 2009). Two studies utilized neuronavigation (Jay et al., 2014; Kuppuswamy et al., 2008) in which structural MRI images assist in determining optimal TMS treatment target location. We did not, however, locate any studies assessing EDA changes paired with neuronavigation to define precise targeting during TMS stimulus-related measurements of EDA responses.

3.3. Brain regions targeted

3.3.1. Dorsolateral prefrontal cortex

The dlPFC was the most targeted brain region across all studies. Casanova et al. applied inhibitory TMS of 0.5 Hz to the right dlPFC, left dlPFC, and bilateral dlPFC, in children with ASD, each for six sessions (Casanova et al., 2014). Over the course of an 18-week monitoring period they observed a significant reduction in EDA and an increase in HRV interpreted as decreased SNS activity and increased parasympathetic nervous system (PNS) activity, respectively (Casanova et al., 2014). In a subsequent study, this group saw decreases in sympathetic arousal reflected in decreased EDA when compared to patients' baselines after applying 6 sessions of LFTMS to the left and then to the right dlPFC over a 12 week period (Wang et al., 2016). Both studies (Casanova et al., 2014; Wang et al., 2016) saw a direct correlation between EDA levels and reduced repetitive behavior scale ratings, post-TMS treatment. Also employing LFTMS, Kuppuswamy et al. observed lengthening of EDA response latency induced by random stimulation of the neck following a single 900-pulse session of 1-Hz TMS applied to right dlPFC compared to the same four participants receiving sham TMS (Kuppuswamy et al., 2008).

In contrast, Notzon et al. applied HF (15 Hz), intermittent theta burst (iTBS) or sham iTBS to the left dlPFC to participants with spider phobia and healthy controls (Notzon et al., 2015). They found that iTBS applied to the left dlPFC did not have a significant effect on subjective rating of anxiety or disgust or on EDA levels. The iTBS group with spider phobia had a marginally significant increase in LF and decrease in HF components of HRV during spider scenes only (Notzon et al., 2015).

Berger et al. employed a complex design comparing the effects of a single session of LF vs. HF, sham-controlled TMS applied to the right dlPFC in healthy females who then viewed emotionally-valenced pictures with concurrent measurement of EDA and HR (Berger et al., 2017). During the presentation of salient pictures, independent of TMS type, subjects had increased EDA and phasic deceleration in HR (PCR), reflective of stimulus-directed attention (Berger et al., 2017). Increase in SC (presumed SNS activation) is thought to represent the initial attention capture (orienting response), whereas the PCR (presumed PNS activation) reflects continuing sensory processing. The presentation of highly arousing negative pictures was accompanied by the most pronounced PCR and the highest EDA while positive pictures elicited the lowest reactions (Berger et al., 2017). A three-way interaction between picture category, TMS stimulation, and TMS type was observed. This interaction appears to be driven by the fact that PCR after 1 Hz TMS was more pronounced in response to high and low arousal negative and neutral pictures compared to positive pictures (Berger et al., 2017). When 10 Hz was applied to the right dlPFC, less difference in HR was seen between the different emotional categories (Berger et al., 2017), corresponding with prior studies showing improved cognitive control in anxiety disorders after HFTMS to the left dlPFC (Berger et al., 2017; Paes et al., 2013; Rossi et al., 2006). Despite these trends in arousal, Berger et al. did not see statistically significant changes in EDA after TMS treatments were completed (Berger et al., 2017). These findings

may support the complexity of ANS response to provoking factors through both SNS involvement in determining initial emotional valence and PNS contribution to maintaining continuing monitoring and response to the stimulus.

3.3.2. Medial prefrontal cortex

Guhn et al. used HF (10 Hz) TMS over the medial prefrontal cortex (mPFC) in healthy, TMS-naïve participants prior to the extinction component of a fear conditioning and extinction paradigm in an attempt to modulate the processing of conditioned fear (Guhn et al., 2014). The TMS group demonstrated a reduced fear response through decreased measurements of fear-potentiated startle reflex and EDA responses compared to sham group. Based upon these results, the authors speculate that mPFC TMS might prove beneficial for anxiety patients when combined with exposure therapy (Guhn et al., 2014).

3.3.3. Primary motor cortex

Bracco et al. applied excitatory and inhibitory TMS to both the right and left M1 in healthy right-handed adults to determine its influence on EDA. It was determined that, in the right M1, excitatory TMS directly correlated with sympathetic activation, preceding increases in EDA, whereas inhibitory TMS inversely correlated with sympathetic activation (Bracco et al., 2017). The authors speculate that GABA-B activity is likely mediated in the M1 and that this could mediate anxiety (Bracco et al., 2017). In the left M1, however, both excitatory and inhibitory frequencies resulted in lower sympathetic activation, preceding lower EDA levels during both trials (Bracco et al., 2017). Overall, these findings suggest that the right hemisphere plays a more specific role in the SNS, as supported by prior literature (Bracco et al., 2017; Critchley, 2002; Zahn et al., 1999a; Zoccolotti et al., 1982).

3.3.4. Supplementary motor cortex

Rodigari and Oliveri (Rodigari and Oliveri, 2014) applied TMS to the supplementary motor cortex (SMA) at 1 Hz during the performance of a task requiring young women to judge the valence and arousal of emotional images (e.g., negative-violent, neutral, and positive-erotic). SMA inhibition resulted in a statistically significant increase in EDA and an increase in emotional response to negative visual stimuli above both baseline responses and occipital TMS, which had no effect on judgments. This implies that the SMA is involved in emotional processing. These findings are consistent with a prior fMRI study showing SMA connections to the frontal, temporal, and cingulate regions in response to emotionally laden words (Citron et al., 2014). Altogether, these findings suggest that the SMA could be an additional target in TMS treatments tracking EDA within a paradigm broadly focused upon emotion regulation or attention bias modification.

3.3.5. Temporoparietal lobe junction

Tsang et al. used a burst firing pattern of ultra-low intensity (1 microTesla), LF (0.5 Hz) TMS applied to the temporoparietal lobe junction (TPJ) in healthy adults, ages 20–25 (Tsang et al., 2009). Subjective improvements in mood and fatigue were noted over the 3-week period when compared to the sham TMS treatments (Tsang et al., 2009). However, no statistically significant changes in HRV or EDA were noted before and after treatments, indicating that the TPJ may not play a role in SNS activity, although it is also possible that this intensity and or pattern of stimulation may not be physiologically potent (Tsang et al., 2009). Jay et al. applied LF (1 Hz) TMS to the right TPJ in patients with DPD, noting that patients reported significant decreases in DPD symptoms, discerned via lower scores on the Cambridge Depersonalization Scale State Version (CDS-S) (Sierra and Berrios, 2000) as compared to their individual baseline scores and to the controls' (Jay et al., 2014). However, no changes in event-related EDA fluctuations or maximum EDA levels were noted, supporting the notion that the TPJ is not important for altering SNS activity and does not influence EDA levels (Jay et al., 2014).

3.3.6. Ventrolateral prefrontal cortex

Jay et al. targeted the right ventrolateral prefrontal cortex (vlPFC) in patients diagnosed with DPD using 1 Hz TMS (2014). They saw that one 15-minute TMS session to the right vlPFC caused increased EDA in the maximum SC response of the DPD group with no change seen in healthy controls (Jay et al., 2014). They also found a trend of increase in spontaneous fluctuations in the treatment group after one treatment session (Jay et al., 2014), indicating that applying LFTMS to vlPFC increases sympathetic activity in patients with DPD, but not in controls. Patients also noted a significant decrease in overall depersonalization symptoms, indicated by lower scores on the CDS-S, as well as an increase in emotional response to stimuli (Jay et al., 2014). A prior study showed that DPD patients in remission had increases in spontaneous EDA fluctuations (Lader and Wing, 1966).

3.4. Correlations with other measures of ANS activity

Many of the studies included other forms of autonomic measurement in addition to EDA. A study done by Schumann et al. assessed different physiologic measures in unmedicated patients with MDD, such as blood pressure variability (BPV), respiratory rate (RR), baroreflex sensitivity, pupillary unrest indices (PUIs), and HRV, in addition to EDA (Schumann et al., 2017). They found that BPV and RR had no statistically significant difference between patients with MDD and the controls (Schumann et al., 2017). However, patients with MDD had decreased baroreflex function and an increase in mean pupillary diameter and in PUIs in both eyes (Schumann et al., 2017). Additionally, patients with MDD displayed statistically significant increases in EDA levels compared to control groups (Schumann et al., 2017).

The most common ANS measures paired with EDA included HRV and patient-reported scales. Casanova et al. analyzed the autonomic dysfunction linked to ASD, manifested as increased basal SNS tone and decreased baseline PNS activity, consistent with previous studies indicating that low HRV is reflective of reduced psychophysiological flexibility in children with ASD (Casanova et al., 2014). After LFTMS to the bilateral dlPFC, patients with ASD experienced less EDA variation and increased HRV in response to auditory or visual triggers as well as a decrease in hyperactivity, indicating that autonomic metrics can potentially serve as a marker for treatment efficacy (Casanova et al., 2014; Wang et al., 2016). Berger et al. found that LFTMS of the right dlPFC results in increased PNS activity while viewing negative emotional stimuli as seen through reduced HRV between pre-emotional and post-emotional trigger (Berger et al., 2017). In addition, when salient visual stimuli were shown (without TMS treatments), the ANS reacted with increased SC and PCR, showing stimulus directed attention (Berger et al., 2017). However, Berger et al. did not find differences in SC after TMS treatments (Berger et al., 2017). Explicit SNS decrease may not have been shown through EDA, but signs of better ANS regulations through increased PNS activity shown by HRV changes were displayed, demonstrating the usefulness of pairing these two autonomic metrics for a more comprehensive assessment of the ANS.

4. Discussion

Our review identified a modest number of studies employing EDA as a measure of ANS activity in response to TMS stimulation. A further limitation is that the majority of studies included only healthy controls and very few clinical studies were relevant to modern treatment paradigms for TMS, i.e., TMS of the dlPFC in patients with MDD. A range of cortical regions were targeted in the literature, including dlPFC, vlPFC, mPFC, M1, SMA and TPJ. Multiple studies reported significant changes in EDA measures, many supported by TMS sham-control. As might be predicted by the widespread cortical, subcortical, excitatory, and inhibitory influences on EDA, EDA response varied depending upon the type and target of TMS stimulation, indicating EDA's plasticity in response to changing conditions. Most studies measured EDA only in

close temporal proximity to TMS delivery, leaving uncertain the durability of the TMS effect. However, TMS is often done in multiple sessions over the course of weeks, which provides the opportunity to assess for changes in baseline EDA, HRV, and other pertinent measures in the pre-TMS recording that is already done at each session (Casanova et al., 2014; Wang et al., 2016).

Depending on the stimulation paradigm used, TMS induces excitatory or inhibitory cortical activity in a circumscribed area under the coil (most commonly the dlPFC) and secondarily influences other parts of the brain that are functionally connected. Therefore, the effect of TMS cortical stimulation on EDA is a function of the stimulated region, the secondary effect of that stimulation on interconnected brain areas, and the presumed tertiary effects on the SNS. The value of EDA as a physiologic correlate of antidepressant or other clinical TMS effect relies on the assumption of a direction of association with clinical response and on reliable and specific effects of TMS stimulation, as well as attention to controlling confounds and experimental conditions.

There is evidence that TMS has indirect effects on the PNS based upon the observation that HR is reduced with stimulation of the dlPFC. HR decelerations are rapidly seen with inhibitory stimulation of the right dlPFC, and excitatory stimulation of left dlPFC may produce this effect as well (Kaur et al., 2020). At a basic level, modulation of the CAN resulting in diminished SNS activity reflected in reduction in EDA may be presumed to result from stimulation of the dlPFC-anterior cingulate pathway. The degree of correlation between nodes in this pathway has been posited to account for differences in antidepressant response amongst dlPFC targets (Fox et al., 2012). This line of inquiry is summarized by a frontovagal network theory and is the basis for a novel method of TMS targeting (Iseger et al., 2020). While the effect of theta burst TMS on HR has been replicated, one study indicated that these effects occurred independent of treatment outcome (Alario et al., 2023).

Additionally, there are studies that support cumulative effects of TMS stimulation on the ANS as measured by HRV (Makovac et al., 2017). We have demonstrated clinically significant improvements in HRV both acutely and longitudinally over the course of TMS therapy (Onyilo et al., 2022). It remains to be shown how EDA compares with other measures of ANS activity across conditions and in response to neuromodulation (McCall et al., 2021). Notably, there are currently no studies comparing changes in pupillary light response (PRL) to EDA changes during TMS treatments. The work of Guhn et al. showing modulation of conditioned fear responses simultaneous with changes in EDA induced by active but not by sham TMS stimulation of the ventromedial PFC is a promising illustration of the potential (Guhn et al., 2014). A study assessing changes in PRL, EDA, and HRV while applying TMS to various cortical targets could be useful in unraveling the complexity of the dorsal nodes of the CAN.

Multiple studies provide evidence that inhibitory TMS applied to the dlPFC, especially on the right hemisphere, decreases EDA levels (Berger et al., 2017; Casanova et al., 2014; Kuppuswamy et al., 2008; Wang et al., 2016). Several patient groups not only had lower EDA levels after treatment but also saw improvements in patient-reported scales, such as improved agitation in patients with ASD, illustrating its potential to predict clinically relevant endpoints (Casanova et al., 2014; Wang et al., 2016). These results are consistent with the results from Kuppuswamy et al. (2008), where LF (1 Hz) TMS was applied to right dlPFC in 4 healthy volunteers, which resulted in increased latency in EDA compared to patient baseline and to sham controls after one session of treatment. LFTMS applied to the vlPFC, M1, and mPFC were also noted to cause changes in EDA, although their results have not been replicated (Bracco et al., 2017; Guhn et al., 2014; Jay et al., 2014). Neither study on the TPJ found statistically significant changes in EDA levels when applying LFTMS in either healthy controls (Tsang et al., 2009) or patients with DPD (Jay et al., 2014). Tsang et al.'s study used a low magnetic field intensity of 1 microTesla for TMS, finding no significant EDA changes and indicating that a magnetic field intensity threshold might need to be met in order to see EDA changes (Tsang et al., 2009).

Dose-response studies are required to determine if the magnetic field intensity threshold that elicits an EDA response is similar to the threshold that elicits a clinical antidepressant response to TMS.

Only one of our included studies assessed patients with MDD, although HFTMS to the left dlPFC has been approved by the FDA for refractory depression treatment since 2008. A model incorporating HR, PUI, and EDA level distinguished MDD from healthy controls with a specificity of 93% and a sensitivity of 83% (Schumann et al., 2017). This suggests the value of a multi-modal measurement of ANS at baseline in MDD patients receiving TMS treatments. Similarly, dominant EDA levels on the right wrist have been associated with worsening depression, further suggestive of the need to consider hemispheric asymmetries in terms of both TMS targeting and EDA response (Fedor et al., 2016). A recent review found that EDA has merit in determining a patient's suicide risk (Wincewicz-Cichecka and Nasierowski, 2020), noticing that different studies observed higher or lower EDA levels in association with increased suicide risk depending on the patient's primary diagnosis. EDA assessments, likely combined with other ANS measures, could provide a personalized approach to suicide risk assessment. These patients could receive adjustments in the treatment plan, including acceleration of TMS therapy or reconsideration of targeting, better leveraging TMS demonstrated potential to reduce suicide risk (Bozzay et al., 2020).

EDA has yet to be assessed as a metric for guiding TMS treatment targeting to the dlPFC. The dlPFC plays an established role in SNS functioning (Beissner et al., 2013). This claim is supported by increased BP following transcranial alternating current (tcACS) over both the right and left dlPFC (Sesa-Ashton et al., 2022). Additionally, EDA changes have been shown to be a reliable metric of sympathetic response (Ghiassi et al., 2020). A study assessing EDA changes by confirming TMS probe placement over the dlPFC using neuronavigation would provide valuable confirmation about location targeting for TMS treatments to be used in standard practice. Future studies could even be done assessing EDA changes moving from the dlPFC toward M1, especially on the right hemisphere, to determine if instantaneous EDA changes can be used to determine which brain region is being targeted by the TMS apparatus (Bracco et al., 2017).

Physiologic measures are of perennial interest in the field of psychiatry, and these studies have demonstrated that EDA can be measured in reliable forms, with a sizeable signal to noise ratio, demonstrating the potential to be a valuable TMS treatment-associated assessment. However, these studies have several limitations. Many of the studies testing patients with a diagnosis did not use a healthy control group (Casanova et al., 2014; Fedor et al., 2016; Wang et al., 2016), and a placebo, or sham TMS, was not always used (Bracco et al., 2017; Casanova et al., 2014; Fedor et al., 2016; Jay et al., 2014; Rodigari and Oliveri, 2014; Wang et al., 2016). Several studies utilized sample sizes that are too small to allow for generalizability (Bracco et al., 2017; Casanova et al., 2014; Fedor et al., 2016; Jay et al., 2014; Kuppuswamy et al., 2008; Rodigari and Oliveri, 2014). Only three studies included another metric of ANS activity (Berger et al., 2017; Casanova et al., 2014; Wang et al., 2016), using HRV alongside EDA. Future studies should attempt to recruit larger sample sizes, utilize sham TMS, and use a healthy control group to increase methodological rigor, and to employ concurrent ANS activity measurements to enhance conclusions drawn from the data.

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Olivia D. Cox: Conceptualization, Methodology, Validation, Investigation, Resources, Data curation, Writing – original draft, Writing – review & editing, Visualization, Supervision, Project administration.
Ananya Munjal: Conceptualization, Methodology, Validation,

Investigation, Resources, Data curation, Writing – original draft, Writing – review & editing, Visualization, Supervision. **William V. McCall:** Validation, Resources, Writing – review & editing. **Brian J. Miller:** Validation, Resources, Writing – review & editing, Visualization. **Chris Baeken:** Validation, Resources, Writing – review & editing. **Peter B. Rosenquist:** Conceptualization, Methodology, Validation, Investigation, Resources, Data curation, Writing – original draft, Writing – review & editing, Visualization, Supervision, Project administration, Funding acquisition.

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References

- Alario, A.A., Pace, B.D., Niciu, M.J., Trapp, N.T., 2023. Transcranial magnetic stimulation induces heart rate decelerations independent of treatment outcome. *Brain Stimul.* 16, 1044–1046.
- Beissner, F., Meissner, K., Bär, K.J., Napadow, V., 2013. The autonomic brain: an activation likelihood estimation meta-analysis for central processing of autonomic function. *J. Neurosci.* 33, 10503–10511.
- Benarroch, E.E., 1993. The central autonomic network: functional organization, dysfunction, and perspective. *Mayo Clin. Proc.* 68, 988–1001.
- Berger, C., Domes, G., Balschat, J., Thome, J., Höppner, J., 2017. Effects of prefrontal rTMS on autonomic reactions to affective pictures. *J. Neural Transm.* 124, 139–152.
- Boucsein, W., 2012. *Electrodermal Activity*. Springer Science & Business Media.
- Bozzay, M.L., Primack, J., Barredo, J., Philip, N.S., 2020. Transcranial magnetic stimulation to reduce suicidality—a review and naturalistic outcomes. *J. Psychiatr. Res.* 125, 106–112.
- Bracco, M., Turriziani, P., Smirni, D., Mangano, R.G., Oliveri, M., 2017. Relationship between physiological excitatory and inhibitory measures of excitability in the left vs. right human motor cortex and peripheral electrodermal activity. *Neurosci. Lett.* 641, 45–50.
- Casanova, M.F., Hensley, M.K., Sokhadze, E.M., El-Baz, A.S., Wang, Y., Li, X., Sears, L., 2014. Effects of weekly low-frequency rTMS on autonomic measures in children with autism spectrum disorder. *Front. Hum. Neurosci.* 8, 851.
- Citron, F.M., Gray, M.A., Critchley, H.D., Weekes, B.S., Ferstl, E.C., 2014. Emotional valence and arousal affect reading in an interactive way: neuroimaging evidence for an approach-withdrawal framework. *Neuropsychologia* 56, 79–89.
- Critchley, H.D., 2002. Electrodermal responses: what happens in the brain. *Neuroscientist* 8, 132–142.
- Dawson, M.E.S.A.M., Filion, D.L., 2016. The electrodermal system. In: Cacioppo, J., Tassinary, L., Berntson, G. (Eds.), *Handbook of Psychophysiology*. Cambridge University Press, Cambridge.
- Fedor, S., Chau, P., Bruno, N., Picard, R.W., Camprodon, J., 2016. Can we predict depression from the asymmetry of electrodermal activity? *Iproceedings* 2, e6117.
- Fox, M.D., Buckner, R.L., White, M.P., Greicius, M.D., Pascual-Leone, A., 2012. Efficacy of transcranial magnetic stimulation targets for depression is related to intrinsic functional connectivity with the subgenual cingulate. *Biol. Psychiatry* 72, 595–603.
- Fredrikson, M., Furmark, T., Olsson, M.T., Fischer, H., Andersson, J., Langstrom, B., 1998. Functional neuroanatomical correlates of electrodermal activity: a positron emission tomographic study. *Psychophysiology* 35, 179–185.
- Ghiassi, S., Greco, A., Barbieri, R., Scilingo, E.P., Valenza, G., 2020. Assessing autonomic function from electrodermal activity and heart rate variability during cold-pressor test and emotional challenge. *Sci. Rep.* 10, 1–13.
- Grace, A.A., 1991. Phasic versus tonic dopamine release and the modulation of dopamine system responsivity: a hypothesis for the etiology of schizophrenia. *Neuroscience* 41, 1–24.
- Guhn, A., Dresler, T., Andreatta, M., Müller, L.D., Hahn, T., Tupak, S.V., Polak, T., Deckert, J., Herrmann, M.J., 2014. Medial prefrontal cortex stimulation modulates the processing of conditioned fear. *Front. Behav. Neurosci.* 8, 44.
- Iseger, T.A., van Bueren, N.E.R., Kenemans, J.L., Gevirts, R., Arns, M., 2020. A frontal-vagal network theory for major depressive disorder: implications for optimizing neuromodulation techniques. *Brain Stimul.* 13, 1–9.
- Jay, E.L., Sierra, M., Van den Eynde, F., Rothwell, J.C., David, A.S., 2014. Testing a neurobiological model of depersonalization disorder using repetitive transcranial magnetic stimulation. *Brain Stimul.* 7, 252–259.

- Kaur, M., Michael, J.A., Hoy, K.E., Fitzgibbon, B.M., Ross, M.S., Iseger, T.A., Arns, M., Hudaib, A.R., Fitzgerald, P.B., 2020. Investigating high- and low-frequency neuro-cardiac-guided TMS for probing the frontal vagal pathway. *Brain Stimul.* 13, 931–938.
- Kuppuswamy, A., Nicotra, A., King, N.K., Ellaway, P., 2008. MO21 Contribution of the cortex to sympathetic skin response studied using image-guided repetitive transcranial magnetic stimulation (rTMS) in healthy individuals. *Clin. Neurophysiol.* S11.
- Lader M.H., Wing L., 1966. *Physiological measures, Sedative drugs, and Morbid Anxiety.* London.
- Makovac, E., Thayer, J.F., Ottaviani, C., 2017. A meta-analysis of non-invasive brain stimulation and autonomic functioning: implications for brain-heart pathways to cardiovascular disease. *Neurosci. Biobehav. Rev.* 74, 330–341.
- Mangina, C.A., Beuzeron-Mangina, J.H., 1996. Direct electrical stimulation of specific human brain structures and bilateral electrodermal activity. *Int. J. Psychophysiol.* 22, 1–8.
- McCall, W.V., Sareddy, S., Youssef, N.A., Miller, B.J., Rosenquist, P.B., 2021. The pupillary light reflex as a point-of-care test for suicide risk: preliminary results. *Psychiatry Res.* 295, 113582.
- Notzon, S., Deppermann, S., Fallgatter, A., Diemer, J., Kroczeck, A., Domschke, K., Zwanzger, P., Ehls, A.C., 2015. Psychophysiological effects of an iTBS modulated virtual reality challenge including participants with spider phobia. *Biol. Psychol.* 112, 66–76.
- O'Reardon, J.P., Solvason, H.B., Janicak, P.G., Sampson, S., Isenberg, K.E., Nahas, Z., McDonald, W.M., Avery, D., Fitzgerald, P.B., Loo, C., 2007. Efficacy and safety of transcranial magnetic stimulation in the acute treatment of major depression: a multisite randomized controlled trial. *Biol. Psychiatry* 62, 1208–1216.
- Onyilo, C.V., McCall, W.V., Rosenquist, P., 2022. Transcranial magnetic stimulation therapy of major depressive disorder improves heart rate variability. *J. ECT* 38, 138–140.
- Paes, F., Baczynski, T., Novaes, F., Marinho, T., Arias-Carrión, O., Budde, H., Sack, A.T., Huston, J.P., Almada, L.F., Carta, M., 2013. Repetitive transcranial magnetic stimulation (rTMS) to treat social anxiety disorder: case reports and a review of the literature. *Clinical practice and epidemiology in mental health.* CP EMH 9, 180.
- Page, M., McKenzie, J., Bossuyt, P., Boutron, I., Hoffmann, T., Mulrow, C., Moher, D., 2021. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *Syst. Rev.* 10 (89), 1–11.
- Pineau, G., Jean, E., Romo, L., Villemain, F., Poupon, D., Gorwood, P., 2022. Skin conductance while facing emotional pictures at day 7 helps predicting antidepressant response at three months in patients with a major depressive episode. *Psychiatry Res.* 309, 114401.
- Posada-Quintero, H.F., Chon, K.H., 2020. Innovations in electrodermal activity data collection and signal processing: a systematic review. *Sensors (Basel)* 20.
- Reutens, S., Nielsen, O., Sachdev, P., 2010. Depersonalization disorder. *Curr. Opin. Psychiatry* 23, 278–283.
- Rodigari, A., Oliveri, M., 2014. Disrupting SMA activity modulates explicit and implicit emotional responses: an rTMS study. *Neurosci. Lett.* 579, 30–34.
- Rossi, S., Cappa, S.F., Ulivelli, M., De Capua, A., Bartalini, S., Rossini, P.M., 2006. rTMS for PTSD: induced merciful oblivion or elimination of abnormal hypermnnesia? *Behav. Neurol.* 17, 195–199.
- Roy, J.C., Leisingertrigona, M.C., Bloch, V., 1977. Reticular thresholds for evoking electrodermal responses in cats. *Arch. Ital. Biol.* 115, 171–184.
- Sarchiapone, M., Gramaglia, C., Iosue, M., Carli, V., Mandelli, L., Serretti, A., Marangon, D., Zeppegno, P., 2018. The association between electrodermal activity (EDA), depression and suicidal behaviour: a systematic review and narrative synthesis. *BMC Psychiatry* 18, 22.
- Schumann, A., Andrack, C., Baer, K.J., 2017. Differences of sympathetic and parasympathetic modulation in major depression. *Progress Neuro-Psychopharmacol. Biological Psychiatry* 79, 324–331.
- Sesa-Ashton, G., Wong, R., McCarthy, B., Datta, S., Henderson, L.A., Dawood, T., Macefield, V.G., 2022. Stimulation of the dorsolateral prefrontal cortex modulates muscle sympathetic nerve activity and blood pressure in humans. *Cerebral Cortex Commun.* 3, tgac017.
- Sierra, M., Berrios, G.E., 2000. The Cambridge Depersonalisation Scale: a new instrument for the measurement of depersonalisation. *Psychiatry Res.* 93, 153–164.
- Sierra, M., Senior, C., Phillips, M.L., David, A.S., 2006. Autonomic response in the perception of disgust and happiness in depersonalization disorder. *Psychiatry Res.* 145, 225–231.
- Thayer, J.F., Lane, R.D., 2000. A model of neurovisceral integration in emotion regulation and dysregulation. *J. Affect. Disord.* 61, 201–216.
- Tranel, D., 2000. Electrodermal activity in cognitive neuroscience: neuroanatomical and neuropsychological correlates. In: Lane, R.D., Nadel, L., Ahern, G.L., Allen, J., Kaszniak, A.W. (Eds.), *Cognitive Neuroscience of Emotion.* Oxford University Press, New York.
- Tranel, D., Damasio, H., 1989. Intact electrodermal skin conductance responses after bilateral amygdala damage. *Neuropsychologia* 27, 381–390.
- Tranel, D., Damasio, H., 1994. Neuroanatomical correlates of electrodermal skin conductance responses. *Psychophysiology* 31, 427–438.
- Tranel, D.T., 1983. The effects of monetary incentive and frustrative nonreward on heart rate and electrodermal activity. *Psychophysiology* 20, 652–657.
- Tsang, E.W., Koren, S.A., Persinger, M.A., 2009. Specific patterns of weak (1 microTesla) transcranial complex magnetic fields differentially affect depression, fatigue, and confusion in normal volunteers. *Electromagn. Biol. Med.* 28, 365–373.
- Wang, Y., Hensley, M.K., Tasman, A., Sears, L., Casanova, M.F., Sokhadze, E.M., 2016. Heart rate variability and skin conductance during repetitive TMS course in children with autism. *Appl. Psychophysiol. Biofeedback* 41, 47–60.
- Winciewicz-Cichecka, K., Nasierowski, T., 2020. Electrodermal activity and suicide risk assessment in patients with affective disorders. *Psychiatr. Pol.* 54, 1137–1147.
- Zahn, T.P., Grafman, J., Tranel, D., 1999a. Frontal lobe lesions and electrodermal activity: effects of significance. *Neuropsychologia* 37, 1227–1241.
- Zahn, T.P., Grafman, J., Tranel, D., 1999b. Frontal lobe lesions and electrodermal activity: effects of significance. *Neuropsychologia* 37, 1227–1241.
- Zoccolotti, P., Scabini, D., Violani, C., 1982. Electrodermal responses in patients with unilateral brain damage. *J. Clin. Exp. Neuropsychol.* 4, 143–150.